

IN4MATX 232: Research in HCI

Class 4:
HCI Research Process

Daniel Epstein

Outline

- How does HCI work get done?
 - Funding
 - Review process
 - Methods
- HCI work inside and outside of CHI
 - Venue “rankings”
 - Conferences vs. Journals
 - Workshops, extended abstracts, etc.

How does HCI work get done?

How does HCI work get done?

Funding

- HCI work is often grant-funded
- In the US, National Science Foundation offers a lot of support
 - Human-Centered Computing (HCC) directive
 - A little of Secure and Trustworthy Cyberspace (SaTC)
- Some support through the National Institutes of Health
 - NIH model often doesn't align super well, less student support and more specific deliverables. But it can work sometimes

How does HCI work get done?

Funding

- Some industry support
 - FAANG-type companies, but some others too
 - Tends to lean a bit more technical, focused on innovation
- Some foundation support
 - Tends to focus on public good, informing policy
 - Can sometimes be hard to align with PhD deliverables
- Support differs a lot from research lab to research lab

Review process

How does HCI work get done?

Review Process

- In CHI, recent switch to Revise & Resubmit (R&R)
- Timeline
 - Initial submission due ~mid-Sept
 - Reviews returned ~mid-Nov
 - Revisions submitted ~mid-Dec
 - Decision released ~mid-Jan

How does HCI work get done?

Review Process

- Two researchers (ACs) will receive your paper
 - 1 primary AC (1AC), 1 secondary AC (2AC)
 - 1AC recruits two “external” reviewers from the research community (not ACs)
 - You receive three full reviews (2AC, 2 external) and a summary (written by 1AC)
- This process ^ is pretty consistent among HCI venues
- In CHI, decision is either reject or invitation to R&R (no direct accept)

How does HCI work get done?

Review Process

The following fields are required by the early submission deadline (September 10). Their content may not be edited after that deadline.

Title

Cut and paste the title from your source file (not from your PDF) into this field. Please Be Sure That Each Important Word in Your Title (on the PDF and in this field) Starts with a Capital Letter Like This Text.

Authors

Enter the authors of the submission below, in the order that they appear on the paper. The contact person does not have to appear in the first position.

Please note that you will need to link each author below to a PCS account. Enter a name or email address and the system will give you a drop-down list of corresponding PCS accounts to choose from. If you enter data that has no matching PCS account, a new account will be created. Creating a new affiliation will cause a pop-up window to appear, so **please make sure your browser is set to allow pop-up windows from PCS.**

Name	Affiliation	Sec
	▼	

[add one](#) | [remove last](#)

Abstract

Copy your submission's abstract into the box below. The abstract should contain a maximum of 150 words.

How does HCI work get done?

Review Process

Primary Subcommittee Selection

Please choose the subcommittee that should review your submission. Your choice influences what type of contribution reviewers will look for, as explained on the [“Selecting a Subcommittee”](#) page.

- ☐ Accessibility and Aging
- ☐ Blending Interaction: Engineering Interactive Systems & Tools
- ☐ Building Devices: Hardware, Materials, and Fabrication
- ☐ Computational Interaction
- ☐ Critical and Sustainable Computing
- ☐ Design
- ☐ Games and Play
- ☐ Health
- ☐ Interacting with Devices: Interaction Techniques & Modalities
- ☐ Interaction Beyond the Individual
- ☐ Learning, Education and Families
- ☐ Privacy and Security
- ☐ Specific Application Areas
- ☐ Understanding People: Quantitative Methods
- ☐ Understanding People: Qualitative Methods
- ☐ Understanding People: Mixed and Alternative Methods
- ☐ User Experience and Usability
- ☐ Visualization

Accessibility and Aging

This subcommittee is suitable for contributions related to the design or study of technology for people with disabilities and/or older adults. Accessibility papers are those that deal with technology designed for or used by people with disabilities including sensory, motor, mobility, and intellectual or learning disabilities. Aging papers are broadly categorized as those dealing with technology designed for or used by people in the later stages of life. Relationships with technology are complex and multifaceted; we welcome contributions across a range of topics aimed at benefiting relevant stakeholder groups and not solely limited to concerns of making technology accessible. Note that if your paper primarily concerns interactions with health data or with healthcare providers, then the [Health](#) subcommittee is probably a better fit, whereas papers reflecting on how technologies are used and/or on designing interfaces and interactions suited to specific needs are a better fit for this subcommittee. We strongly suggest that authors review [this Accessible Writing Guide](#) in order to adopt a writing style that refers to stakeholder groups using appropriate terminology. Submissions to this subcommittee will be evaluated in part based on their inclusion of and potential impact on their target user groups and other stakeholders. This subcommittee balances the rigor required in all CHI submissions with awareness of the challenges of conducting research in these important areas. This subcommittee welcomes all contributions related to accessibility and aging, including empirical, theoretical, conceptual, methodological, design, and systems contributions.

Subcommittee Chairs

- Aisling Kelliher, Virginia Tech, USA
- Kristen Shinohara, Rochester Institute of Technology, USA
- Karyn Moffatt, McGill University, Canada
- Stacey Branham, University of California, USA

How does HCI work get done?

Review Process

Key Words

Choose between two and four key words to describe your submission.

Domain

- ☐ Accessibility
- ☐ Audio/Video
- ☐ Behavior Change
- ☐ Collaboration
- ☐ Commerce / Business
- ☐ Computer Mediated Communication
- ☐ Computer Vision
- ☐ Creativity Support
- ☐ Crisis/Disaster
- ☐ Critical/Activism/Ethics
- ☐ Crowdfunding
- ☐ Crowdsourcing
- ☐ Cultural Heritage/History
- ☐ Design Methods
- ☐ Education/Learning
- ☐ Email/Texting/Communication
- ☐ Embodied Interaction
- ☐ Emotion / Affective Computing
- ☐ Entertainment
- ☐ Fabrication
- ☐ Fashion/Clothing

- ☐ Children/Parents
- ☐ Crowdsourced
- ☐ HCI for Development
- ☐ Individuals with Disabilities & Assistive Technologies
- ☐ Older Adults
- ☐ Teens

Environments

- ☐ Automobile
- ☐ City
- ☐ Home
- ☐ Medical: Nursing Homes/Hospitals
- ☐ Rural Areas
- ☐ Schools/Educational Setting
- ☐ Transportation
- ☐ Workplaces

Devices

- ☐ Ambient Devices / Internet of Things
- ☐ Desktop/Laptop Computers
- ☐ Mobile Devices: Phones/Tablets
- ☐ Public Displays
- ☐ Robot

How does HCI work get done?

Review Process

(Optional) Supplementary Material (in a ZIP file)

Include below any supporting files, such as a video figure or supplementary material (e.g., source code, data, analysis files, etc.). We strongly encourage authors to provide such supplementary material.

Authors providing supplementary material must upload this material in a single ZIP file, and provide descriptive text in the form below. This material will be made available to reviewers.

Note

New file

No file chosen

Description of Supplementary Files

We strongly encourage sharing of supplementary material including source code, data, analysis files, experimental design (anonymized similar to the recommendations for your paper), and then as part of the final submission (if accepted), so that reviewers can view/execute the files, and any other relevant text/information explaining how this material relates to the paper.

If you included a ZIP file of supplementary material, please provide a description here. For example, this description should include a brief description of the files, and any other relevant text/information explaining how this material relates to the paper.

How does HCI work get done?

Review Process

(Optional) External Reviewer Recommendations

You may list up to five external reviewers who *are not members of the program committee* whom you believe [the program committee](#), are not in [conflict](#) with any author of your paper, and are not aware of this submission recommended.

Where does HCI work get done?

Where does HCI work get done

- Yes, the US submits 2x the number of papers of the next highest country (China)
- Large countries are large
- Still, participation is disproportionate
- Lots of initiatives, lots of description of how evaluation criteria favors certain types of scholarship

Qatar	1	3	4	25.00
Romania	2	2	4	50.00
Russia	1	1	2	50.00
Saudi Arabia	1	6	7	14.29
Singapore	21	26	47	44.68
Slovakia	0	1	1	0.00
South Africa	0	2	2	0.00
Spain	1	14	15	6.67
Sri Lanka	1	0	1	100.00
Sweden	19	27	46	41.30
Switzerland	16	55	71	22.54
Taiwan	6	27	33	18.18
Turkey	3	7	10	30.00
Ukraine	0	1	1	0.00
United Kingdom	76	257	333	22.82
United States of America	438	933	1371	31.95
Uruguay	0	1	1	0.00
Vietnam	1	3	4	25.00

Where does HCI work get done

- Some things are definitely easier for US scholars
 - People often say “X% of Americans experience XYZ problem” and nobody criticizes
- Other countries are catching up
 - When I started in HCI, there were almost no papers from China, and significantly fewer from elsewhere in Asia

Chinese Universities Surge in Global Rankings as U.S. Schools Slip

Harvard still dominates, though it fell to No. 3 on a list measuring academic output. Other American universities are falling farther behind their global peers.

[Listen to this article · 13:04 min](#) [Learn more](#) [Share full article](#) [1.3K](#)



HCI work outside of CHI

HCI work outside of CHI

Venue “rankings”

	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	Computer Human Interaction (CHI)	<u>139</u>	185
2.	Proceedings of the ACM on Human-Computer Interaction	<u>88</u>	138
3.	International Journal of Human-Computer Interaction	<u>83</u>	124
4.	Behaviour & Information Technology	<u>73</u>	116
5.	IEEE Transactions on Affective Computing	<u>72</u>	115
6.	International Journal of Human-Computer Studies	<u>64</u>	104
7.	Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies	<u>64</u>	95
8.	Virtual Reality	<u>62</u>	115
9.	International Journal of Interactive Mobile Technologies	<u>60</u>	86
10.	ACM Transactions on Computer-Human Interaction	<u>55</u>	90
11.	ACM Designing Interactive Systems Conference	<u>54</u>	71
12.	ACM Symposium on User Interface Software and Technology	<u>53</u>	79
13.	ACM/IEEE International Conference on Human Robot Interaction	<u>52</u>	78
14.	Frontiers in Virtual Reality	<u>52</u>	76
15.	IEEE Virtual Reality Conference	<u>52</u>	66

HCI work outside of CHI

Venue “rankings”

- What is an h-index?
- Largest number “h” for a venue/person/etc. such that “h” papers have at least “h” citations
- Google’s are calculated by looking at all publications from particular venues in the past five years (h5-index)

HCI work outside of CHI

Venue “rankings”

- There are 139 CHI papers published from 2020-2025 with at least 139 citations
- 88 PACM HCI papers... etc.

	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	Computer Human Interaction (CHI)	<u>139</u>	185
2.	Proceedings of the ACM on Human-Computer Interaction	<u>88</u>	138
3.	International Journal of Human-Computer Interaction	<u>83</u>	124
4.	Behaviour & Information Technology	<u>73</u>	116
5.	IEEE Transactions on Affective Computing	<u>72</u>	115
6.	International Journal of Human-Computer Studies	<u>64</u>	104
7.	Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies	<u>64</u>	95
8.	Virtual Reality	<u>62</u>	115
9.	International Journal of Interactive Mobile Technologies	<u>60</u>	86
10.	ACM Transactions on Computer-Human Interaction	<u>55</u>	90
11.	ACM Designing Interactive Systems Conference	<u>54</u>	71
12.	ACM Symposium on User Interface Software and Technology	<u>53</u>	79
13.	ACM/IEEE International Conference on Human Robot Interaction	<u>52</u>	78
14.	Frontiers in Virtual Reality	<u>52</u>	76
15.	IEEE Virtual Reality Conference	<u>52</u>	66

**Discussion: what does an h-index
measure well? What does it
measure poorly?**

HCI work outside of CHI

Venue “rankings”

- What’s favored in ranking by h-index?
 - Large venues: venues that publish a lot of papers
 - Old venues: any venue less than 5 years old is at a big disadvantage
 - Topical venues: venues doing work which is of interest right now
 - Work where citations are a valuable measure of impact.
Which can exclude policy-influencing work, product-influencing work, etc.
- Rankings are reductive, focus more holistically on the scholarship quality

HCI work outside of CHI

Tenure Evaluation

- At many places, h-index comes up in tenure evaluation
 - I've seen tenure letters explicitly quote h-index and citation count
 - Less so in faculty hiring given shorter records, but venues and # of publications still come up
- Holistic review is better in my opinion, but people look for shorthands and ways to compare across individuals/fields/disciplines

HCI work outside of CHI

Faculty hiring

- Research area matters most.
 - “Who would this person collaborate with? Be on their student’s committees?”
 - “We can’t support the facilities needs of a hardware person”
- h-index is less important in faculty hiring given shorter records
 - But venues, # of publications, and citation counts still come up a lot
- Again, holistic review is better in my opinion, but people look for shorthands
 - We got ~275 applicants for 1 faculty slot this year

HCI work outside of CHI

Venue “rankings”

- I think there’s valuable work being published at all the venues we’ll talk about this quarter, plus a few others
- I know less about the IEEE venues, they tend to be more prominent in Engineering

	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	Computer Human Interaction (CHI)	<u>139</u>	185
2.	Proceedings of the ACM on Human-Computer Interaction	<u>88</u>	138
3.	International Journal of Human-Computer Interaction	<u>83</u>	124
4.	Behaviour & Information Technology	<u>73</u>	116
5.	IEEE Transactions on Affective Computing	<u>72</u>	115
6.	International Journal of Human-Computer Studies	<u>64</u>	104
7.	Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies	<u>64</u>	95
8.	Virtual Reality	<u>62</u>	115
9.	International Journal of Interactive Mobile Technologies	<u>60</u>	86
10.	ACM Transactions on Computer-Human Interaction	<u>55</u>	90
11.	ACM Designing Interactive Systems Conference	<u>54</u>	71
12.	ACM Symposium on User Interface Software and Technology	<u>53</u>	79
13.	ACM/IEEE International Conference on Human Robot Interaction	<u>52</u>	78
14.	Frontiers in Virtual Reality	<u>52</u>	76
15.	IEEE Virtual Reality Conference	<u>52</u>	66

HCI work outside of CHI

Venue “rankings”

- All said, there is some underlying snobbery around CHI as the top HCI venue
- CHI’s website even says that it’s the “leading international conference”

	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	Computer Human Interaction (CHI)	<u>139</u>	185
2.	Proceedings of the ACM on Human-Computer Interaction	<u>88</u>	138
3.	International Journal of Human-Computer Interaction	<u>83</u>	124
4.	Behaviour & Information Technology	<u>73</u>	116
5.	IEEE Transactions on Affective Computing	<u>72</u>	115
6.	International Journal of Human-Computer Studies	<u>64</u>	104
7.	Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies	<u>64</u>	95
8.	Virtual Reality	<u>62</u>	115
9.	International Journal of Interactive Mobile Technologies	<u>60</u>	86
10.	ACM Transactions on Computer-Human Interaction	<u>55</u>	90
11.	ACM Designing Interactive Systems Conference	<u>54</u>	71
12.	ACM Symposium on User Interface Software and Technology	<u>53</u>	79
13.	ACM/IEEE International Conference on Human Robot Interaction	<u>52</u>	78
14.	Frontiers in Virtual Reality	<u>52</u>	76
15.	IEEE Virtual Reality Conference	<u>52</u>	66

Conferences and Journals

HCI work outside of CHI

Conferences vs. Journals

- What's the difference?
 - Conferences: presentation (in-person, virtual, etc.) is a core component
 - Journals: they are published, but not usually presented
- HCI's roots are in a conference tradition
 - Conferences are typical in Computer Science

HCI work outside of CHI

Conferences vs. Journals

- HCI's "conference" publications look like "journal" publications elsewhere
 - Full articles, not abstracts
 - Rigorous peer review
 - Similar acceptance rates*

HCI work outside of CHI

Conferences vs. Journals

- The broader sentiment in academic scholarship is that conferences are less reputable than journals
 - In many conferences, what's accepted is an abstract that's later elaborated on
 - Many conferences do not have formal peer review
- Broader CS has always had an issue with conferences being interpretable
 - E.g., the rest of Science & Engineering doesn't recognize the value

HCI work outside of CHI

Conferences vs. Journals

- This conference/journal tension has only gotten worse as HCI has gotten more interdisciplinary
 - Social sciences are journal-driven, or sometimes larger pieces like books
 - Health is journal-driven
 - ...
- Many universities only consider journals indexed by particular agencies for hiring/promotion/tenure
 - Moreso true outside the US, and there has been some progress

HCI work outside of CHI

Conferences vs. Journals

- So, there's been some move to make HCI publications journal publications
 - CSCW has become Proceedings of the ACM in HCI, CSCW Issue
 - Proceedings of the ACM in HCI publishes other former conferences as well
- CHI is stubborn, but might switch one day

HCI work outside of CHI

Conferences vs. Journals

- Overall, don't worry about it, unless you're going for a very specific kind of job
 - Academic position abroad
 - “The first” HCI hire in an Engineering department
- People are becoming more and more understanding

Making your first HCl contribution

Your first HCI contribution

Short papers

- Very hard to make a short paper contribution
 - They get accepted at a far lower rate
- Why? Reviewers want more detail and depth
 - More methodological details, more engagement with related work, richer discussion
 - Misunderstanding by authors that shorter == lower standard
- But when they do get published, the quality is high
 - A few of my most-cited papers are short papers

Your first HCI contribution

Workshops and Extended Abstracts

- How do I get started doing HCI work? Are there lower-burden ways of dipping my toe into the field?
- Beyond joining existing HCI projects, being mentored by HCI scholars, etc., there are some venues designed for in-progress or less polished work

Your first HCI contribution

Workshops and Extended Abstracts

- Workshops
 - Typically occur in the day or two before/after a conference
 - Usually have a specific topic in mind
- Example workshops from CHI 2025:
 - Generative AI and HCI
 - Future of Interactive Health
 - Advancing Post-Growth HCI

Your first HCI contribution

Workshops and Extended Abstracts

- Workshops
 - Not just at CHI: most of the conferences we will touch on this quarter have workshops
 - Can be a great way to meet people interested in a similar topic
 - Work is non-archival, so you can edit/expand following a workshop

Your first HCI contribution

Workshops and Extended Abstracts

- Extended abstracts
 - Sometimes called “posters” or “late-breaking work” depending on the venue
 - Really just intended to be work that’s not yet at the level of a “full” paper
 - Often still reviewed, albeit with a lower bar
 - Usually comes with presentation at a poster session at the conference
 - Sometimes these are indexed online and can accrue citations, but they’re not really intended to be treated as rigorously reviewed

IN4MATX 232: Research in HCI

Class 4:
HCI Research Process

Daniel Epstein